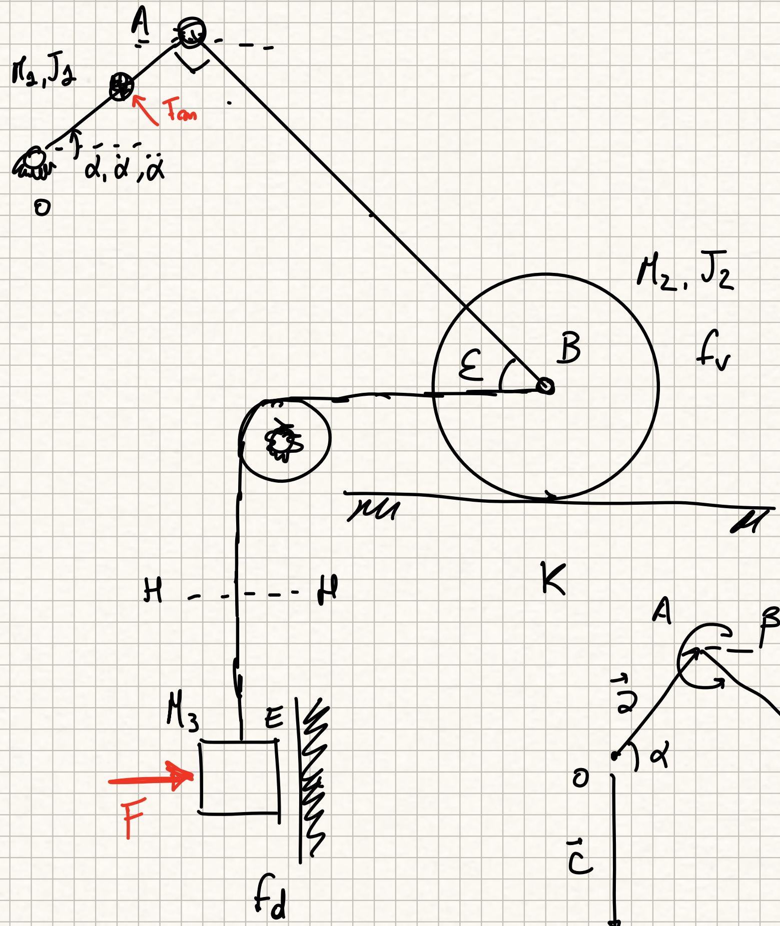
Es 1

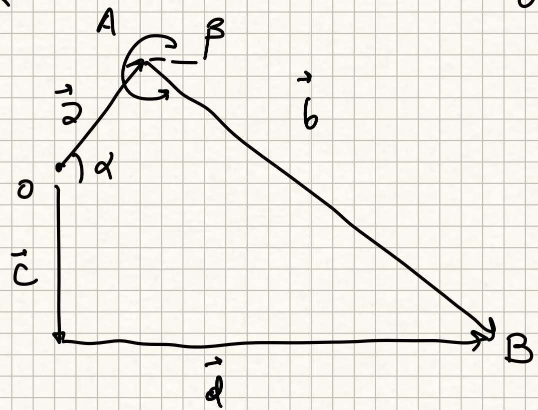


1. v_M, a_M
2. Tiro H-H
3. Reazione asta AB
4. F_m per $\dot\alpha, \ddot\alpha$

$gdl = 15$

$gdv = 2_O + 2_A + 2_B + 2_K + 2_D + 2_E + 2_{FUNE}$

$= 14$ gdv



$\beta = \frac{3\pi}{2} + \alpha$

$\varepsilon = \frac{\pi}{2} - \alpha$

const	a	b	c	γ	δ
var	α	β	d		

$$\begin{cases} a\cos\alpha + b\cos\beta = c\cos\gamma + d\cos\delta \\ a\sin\alpha + b\sin\beta = c\sin\gamma + d\sin\delta \end{cases} \rightarrow \begin{matrix} a\cos\alpha + b\cos\beta = d \\ a\sin\alpha + b\sin\beta = -c \end{matrix}$$

$$\begin{cases} -a\dot\alpha\sin\alpha - b\dot\beta\sin\beta = \dot d\cos\delta \\ a\dot\alpha\cos\alpha + b\dot\beta\cos\beta = \dot d\sin\delta \end{cases} \Rightarrow \begin{bmatrix} b\sin\beta & \cos\delta \\ -b\cos\beta & \sin\delta \end{bmatrix} \begin{Bmatrix} \dot\beta \\ \dot d \end{Bmatrix} = \begin{Bmatrix} -a\dot\alpha\sin\alpha \\ a\dot\alpha\cos\alpha \end{Bmatrix}$$

$\Rightarrow \dot\beta = -0.433$ rad/s

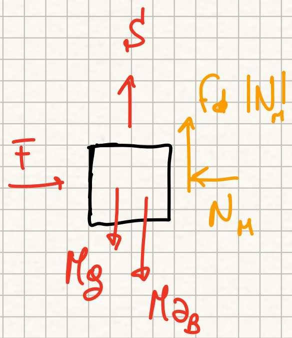
$\dot d = -0.5$ m/s

$$\begin{cases} -a\ddot\alpha\sin\alpha - a\dot\alpha^2\cos\alpha - b\ddot\beta\sin\beta - b\dot\beta^2\cos\beta = \ddot d\cos\delta \\ a\ddot\alpha\cos\alpha - a\dot\alpha^2\sin\alpha + b\ddot\beta\cos\beta - b\dot\beta^2\sin\beta = \ddot d\cos\delta \end{cases} \Rightarrow$$

$\ddot\beta = -1.0658$ rad/s^2

$\ddot d = -1.375$ m/s^2

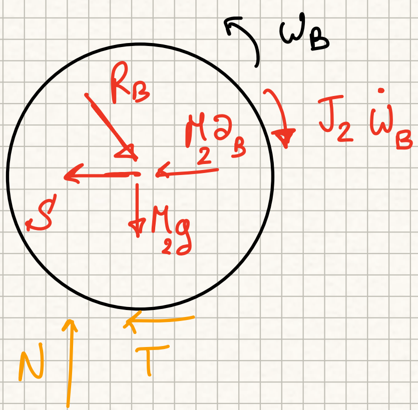
$\vec v_B = \dot d\,\vec i = (-0.5\,\vec i)$ m/s $\qquad \vec a_B = \ddot d\,\vec i = (-1.375\,\vec i)$ m/s^2



$$N_M = F$$

$$S + f_d F - M_3(g + a_B) = 0$$

$$\Rightarrow S = M_3(g + a_B) - f_d F \quad \Rightarrow \quad \boxed{S = 36.18\ N}$$



$$\begin{cases} R_B \cos\varepsilon - M_2 a_B - S - T = 0 \\ -R_B \sin\varepsilon - M_2 g + N = 0 \\ -TR - N f_v R - J_2 \dot{\omega}_B = 0 \end{cases}$$

$$\vec{\omega}_B = -\frac{v_B}{R}\vec{k} = (+1\vec{k})\ \frac{rad}{s} \qquad \dot{\vec{\omega}}_B = -\frac{a_B}{R}\vec{k} = (2.75\vec{k})\ \frac{rad}{s^2}$$

$$R_B = \frac{N - M_2 g}{\sin\varepsilon}, \qquad T = -N f_v - J_2 \frac{\dot{\omega}_B}{R}$$

$$\frac{N - M_2 g}{\sin\varepsilon}\cos\varepsilon - M_2 a_B - S + N f_v + J_2 \frac{\dot{\omega}_B}{R} = 0$$

$$\Rightarrow N = \frac{M_2 g \frac{\cos\varepsilon}{\sin\varepsilon} + M_2 a_B + S - J_2 \frac{\dot{\omega}_B}{R}}{\frac{\cos\varepsilon}{\sin\varepsilon} + f_v} = 51.35\ N$$

$$\boxed{R_B = \frac{N - M_2 g}{\sin\varepsilon} = 47.97\ N}$$

$$\frac{dE_c}{dt} = M_1 v_G a_{Gt} + J_1 \dot{\alpha}\ddot{\alpha} + M_2 v_B a_B + J_2 \dot{\omega}_B \omega_B + M_3 v_B a_B$$

$$v_G = \dot{\alpha}\,\frac{a}{2} = 0.125 \ m/s \,, \qquad a_{at} = \ddot{\alpha}\,\frac{a}{2} = 0.25$$

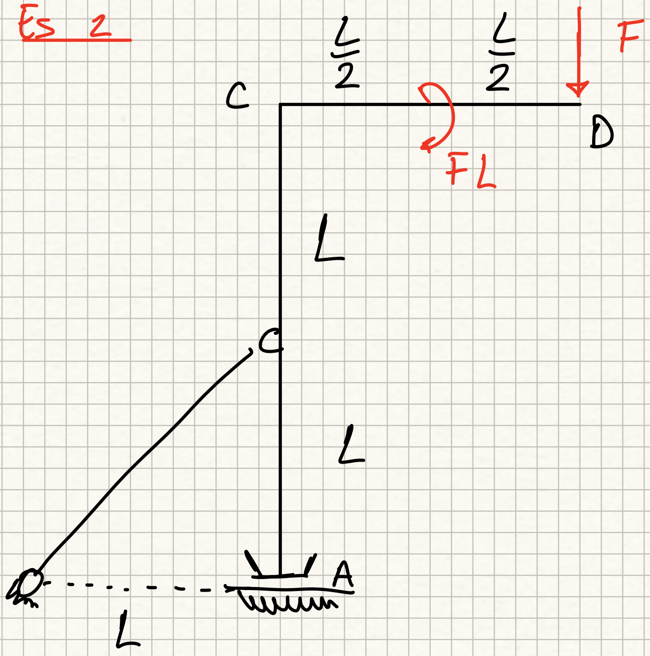
$$\Rightarrow \quad \frac{dE_c}{dt} = 9.2812 \ W$$

$$\Sigma W = -M_1 g\, v_{Gy} - M_3 g\, v_B - N f_v R\, \omega_B - F f_d \,|v_B| + F_m\, \dot{\alpha}\,\frac{a}{2}$$

$$v_{Gy} = \dot{\alpha}\,\frac{a}{2}\cos\alpha = 0.1083 \ m/s$$

$$F_m = \frac{dE_c/dt + M_1 g\, v_{Gy} + M_3 g\, v_B + N f_v R\, \omega_B + F f_d\, v_{Bx}}{\dot{\alpha}\,\frac{a}{2}} \quad \Rightarrow \quad F_m = -79.18 \ N$$

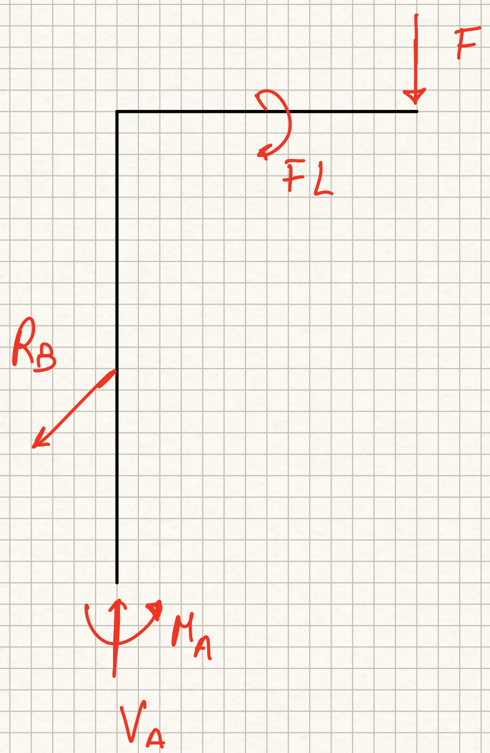
Es 2



? = RV

? = AZIONI ASTA ABCD

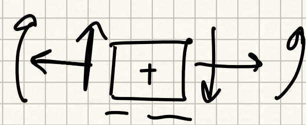
$M = FL$

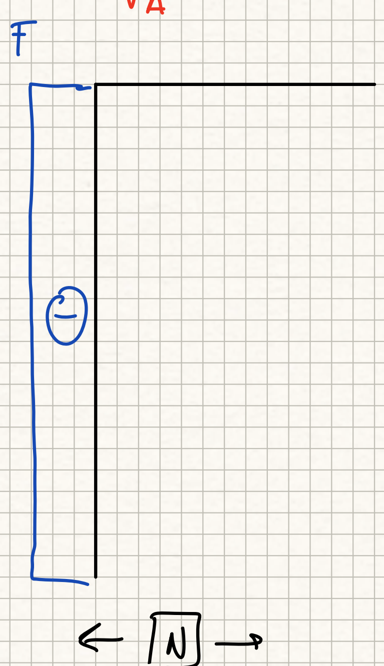


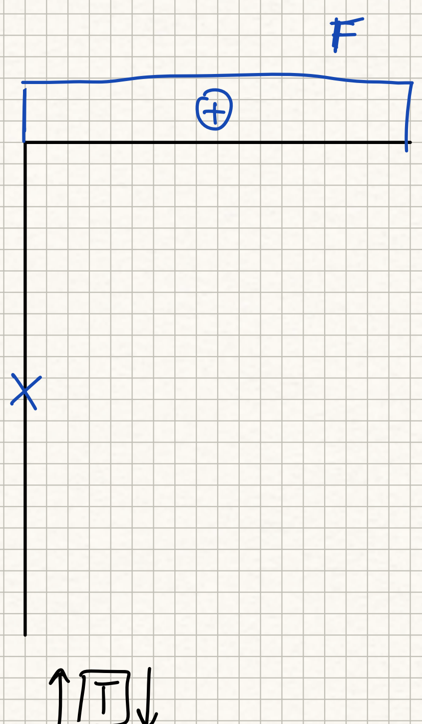
$$M_A - FL - FL = 0 \Rightarrow M_A = 2FL$$

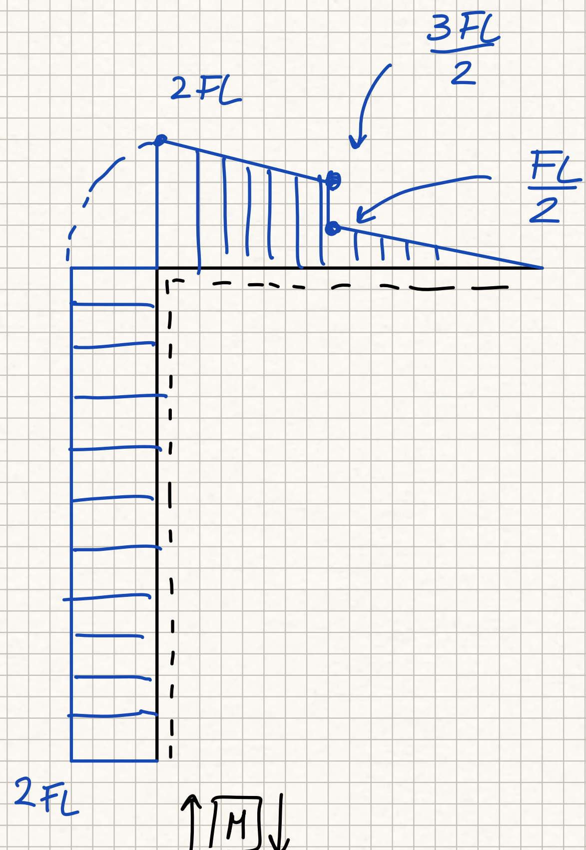
$$R_B = 0$$

$$V_A = F$$







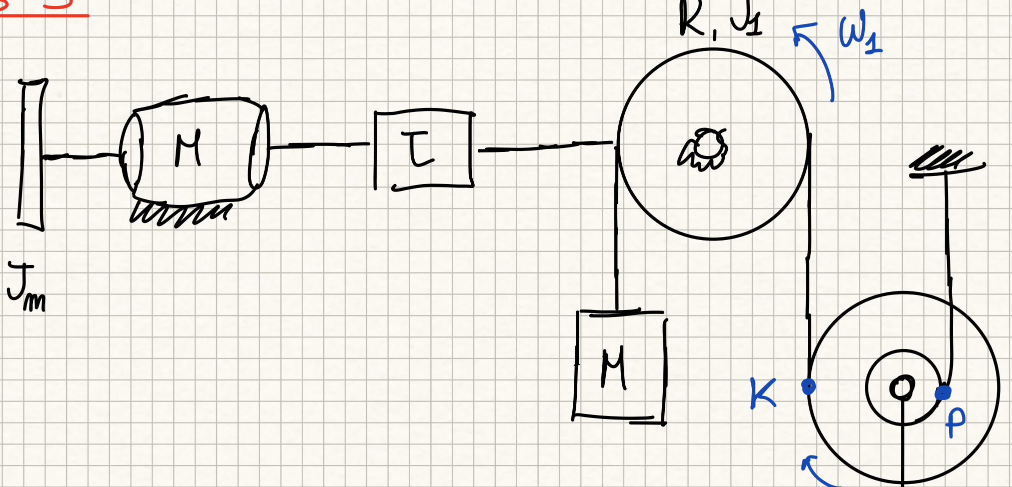


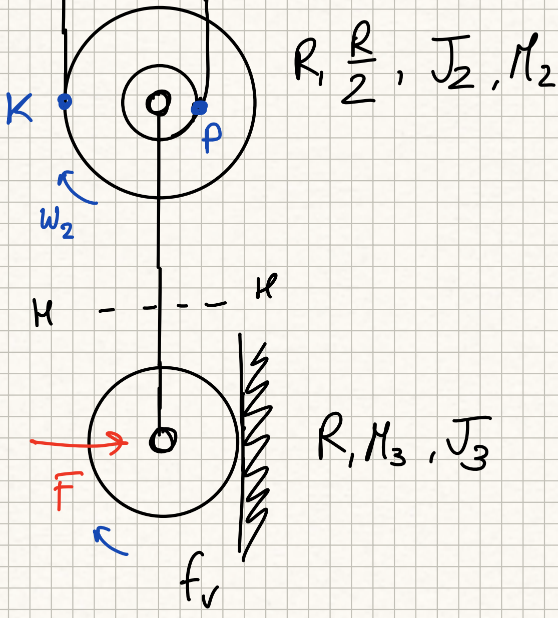
Es 3

① C_m, ω_m @ regime

② a_3 allo spunto

③ verifico di aderenza





$$\tau \omega_m R = \omega_2 \frac{3R}{2} \Rightarrow \omega_2 = \frac{2}{3}\tau\omega_m$$

$$v_2 = \frac{2}{3}\tau\omega_m \frac{R}{2} = \frac{\tau\omega_m R}{3}$$

$$\omega_3 = \frac{\tau\omega_m}{3}$$

$$W_1 = C_m\omega_m - J_m\omega_m\dot{\omega}_m$$

$$\left.\frac{dE_c}{dt}\right|_u = J_1\tau^2\omega_m\dot{\omega}_m + M\tau^2R^2\omega_m\dot{\omega}_m + J_2\frac{4}{9}\tau^2\omega_m\dot{\omega}_m + M_2\frac{\tau^2R^2}{9}\omega_m\dot{\omega}_m +$$
$$+ M_3\frac{\tau^2R^2}{9}\omega_m\dot{\omega}_m + J_3\frac{\tau^2}{9}\omega_m\dot{\omega}_m$$

$$J^* = J_1\tau^2 + M\tau^2R^2 + J_2\frac{4}{9}\tau^2 + M_2\frac{\tau^2R^2}{9} + M_3\frac{\tau^2R^2}{9} + J_3\frac{\tau^2}{9} = 0.0836 \ kg\,m^2$$

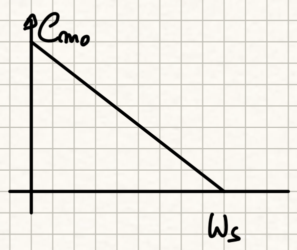
$$\Sigma W_u = Mg\tau\omega_m R - M_2 g\frac{\tau\omega_m R}{3} - M_3 g\frac{\tau\omega_m}{3}R - F f_v R\frac{\tau\omega_m}{3} =$$
$$= \left(Mg\tau R - M_2 g\frac{\tau R}{3} - M_3 g\frac{\tau R}{3} - F f_v R\frac{\tau}{3}\right)\omega_m \rightarrow C_m^* = 3.752 \ Nm$$

① C_m a regime

$W_1 = C_m \omega_m$, $W_2 = C_u^* \omega_m$

$C_u^* > 0 \Rightarrow$ retrogrado $\Rightarrow C_m + \eta_R C_u^* = 0 \Rightarrow C_m = -\eta_R C_u^*$

$C_m = -2.25\ Nm$



$$C_m = C_{m0} - \frac{C_{m0}}{\omega_s}\omega_m$$

$$\Rightarrow \omega_m = \frac{C_{m0} - C_m}{C_{m0}}\omega_s = 278.14\ rad/s$$

② a_3 allo spunto

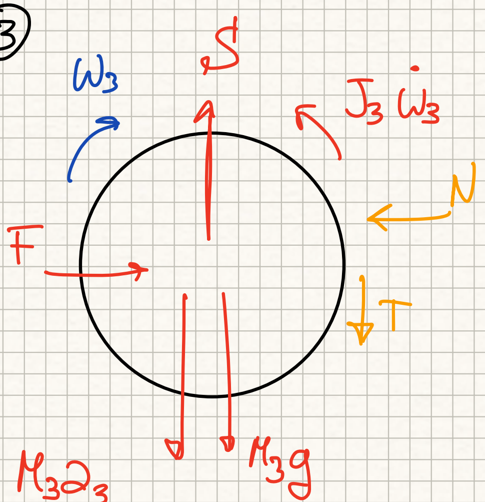
$W_1 = C_m \omega_m - J_m \omega_m \dot{\omega}_m$

$W_2 = \underbrace{C_u^* \omega_m}_{>0} - \underbrace{J_u^* \omega_m \dot{\omega}_m}_{>0}$

IPOTIZZO MOTO RETROGRADO $\Rightarrow C_{m0} - J_m \dot{\omega}_m + \eta_R C_u^* - \eta_R J_u^* \dot{\omega}_m = 0$

$$\dot{\omega}_m = \frac{C_{m0} + \eta_R C_u^*}{J_m + \eta_R J_u^*} = 21.19\ \frac{rad}{s^2} \Rightarrow a_3 = \frac{\tau \dot{\omega}_m R}{3} = 0.3531\ m/s^2$$

③



$N = F$

$TR = J_3 \dot{\omega}_3 + N f_v R \Rightarrow T = \frac{J_3 \dot{\omega}_3}{R} + N f_v$

$$T = \frac{J_3}{R}\frac{\tau \dot{\omega}_m}{3} + F f_v$$

$S = M_3 a_3 + M_3 g + T \Rightarrow S = 23.65\ N$

$\Rightarrow T = 3.325\ N$, $T_{lim} = f_s N = 35\ N$